Lecture 12: The Death of CAPM?

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Intro

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- Last lecture gave us the first cracks in CAPM
- Today we will kill the CAPM and replace it with something else...
- Main theme: size and value move from "anomalies" to new factors
- Momentum gets momentum!
- Hendershott, Livdan, and Rösch (2020): tales of night and day

Flight Plan

- Step 1: why Fama and French (1992) became the "death of CAPM" moment
- Step 2: how Fama and French (1993) turned characteristics into traded factors
- Step 3: momentum and the next challenge (Jegadeesh and Titman 1993; Carhart 1997)
- Step 4: who is awake and trading to generate anomalies? (Hendershott, Livdan, and Rösch 2020)

Part I: The CAPM Break

Fama-French (1992): Question and Setup

- Core question: does market beta explain average returns in the cross-section? (Fama and French 1992)
- Benchmark implication under CAPM: beta should dominate characteristics
- Banz (1981) had already shown the size effect: small firms earn higher returns
- Rosenberg, Reid, and Lanstein (1985) had shown the value effect: high book-to-market firms earn higher returns
- The complicated part is ensuring that single-sort on characteristics do not proxy for β!

FF (1992): Portfolio-Sorting Design

- Each June, they would compute β for each stock using past data
- Form 10 portfolios sorted on deciles of market-cap (size)
- Inside each portfolio create 10 portfolios sorted on β: 100 portfolios in total
- Keep track of returns of each portfolio and redo it next June
- You get 100 time series of portfolio returns
- Compute the β for each portfolio and let $\beta_{i,t} \equiv \beta_{p(i,t)}$
- $p(i,t) \in \{1, 2, 3, \dots, 100\}$: portfolio where stock i belongs at time t

FF (1992): Beta vs Size

	All	Low-β	β-2	β-3	β-4	β-5	β-6	β-7	β-8	β-9	High-β
All	1.25	1.34	1.29	1.36	1.31	1.33	1.28	1.24	1.21	1.25	1.14
Small-ME	1.52	1.71	1.57	1.79	1.61	1.50	1.50	1.37	1.63	1.50	1.42
ME-2	1.29	1.25	1.42	1.36	1.39	1.65	1.61	1.37	1.31	1.34	1.11
ME-3	1.24	1.12	1.31	1.17	1.70	1.29	1.10	1.31	1.36	1.26	0.76
ME-4	1.25	1.27	1.13	1.54	1.06	1.34	1.06	1.41	1.17	1.35	0.98
ME-5	1.29	1.34	1.42	1.39	1.48	1.42	1.18	1.13	1.27	1.18	1.08
ME-6	1.17	1.08	1.53	1.27	1.15	1.20	1.21	1.18	1.04	1.07	1.02
ME-7	1.07	0.95	1.21	1.26	1.09	1.18	1.11	1.24	0.62	1.32	0.76
ME-8	1.10	1.09	1.05	1.37	1.20	1.27	0.98	1.18	1.02	1.01	0.94
ME-9	0.95	0.98	0.88	1.02	1.14	1.07	1.23	0.94	0.82	0.88	0.59
Large-ME	0.89	1.01	0.93	1.10	0.94	0.93	0.89	1.03	0.71	0.74	0.56

FF (1992): Beta vs Book-to-Market Ratio

Portfolio	1A	1B	2	3	4	5	6	7	8	9	10A	10B
Return	0.30	0.67	0.87	0.97	1.04	1.17	1.30	1.44	1.50	1.59	1.92	1.83
β	1.36	1.34	1.32	1.30	1.28	1.27	1.27	1.27	1.27	1.29	1.33	1.35
$\ln(ME)$	4.53	4.67	4.69	4.56	4.47	4.38	4.23	4.06	3.85	3.51	3.06	2.65
$\ln(BE/ME)$	−2.22	−1.51	−1.09	−0.75	−0.51	−0.32	−0.14	0.03	0.21	0.42	0.66	1.02
$\ln(A/ME)$	−1.24	−0.79	−0.40	−0.05	0.20	0.40	0.56	0.71	0.91	1.12	1.35	1.75
$\ln(A/BE)$	0.94	0.71	0.68	0.70	0.71	0.71	0.70	0.68	0.70	0.70	0.70	0.73
E/P dummy	0.29	0.15	0.10	0.08	0.08	0.08	0.09	0.09	0.11	0.15	0.22	0.36
$E(+)/P$	0.03	0.04	0.06	0.08	0.09	0.10	0.11	0.11	0.12	0.12	0.11	0.10
Firms	89	98	209	222	226	230	235	237	239	239	120	117

FF (1992): Size vs BE/ME Interaction

	All	Low	2	3	4	5	6	7	8	9	High
All	1.23	0.64	0.98	1.06	1.17	1.24	1.26	1.39	1.40	1.50	1.63
Small-ME	1.47	0.70	1.14	1.20	1.43	1.56	1.51	1.70	1.71	1.82	1.92
ME-2	1.22	0.43	1.05	0.96	1.19	1.33	1.19	1.58	1.28	1.43	1.79
ME-3	1.22	0.56	0.88	1.23	0.95	1.36	1.30	1.30	1.40	1.54	1.60
ME-4	1.19	0.39	0.72	1.06	1.36	1.13	1.21	1.34	1.59	1.51	1.47
ME-5	1.24	0.88	0.65	1.08	1.47	1.13	1.43	1.44	1.26	1.52	1.49
ME-6	1.15	0.70	0.98	1.14	1.23	0.94	1.27	1.19	1.19	1.24	1.50
ME-7	1.07	0.95	1.00	0.99	0.83	0.99	1.13	0.99	1.16	1.10	1.47
ME-8	1.08	0.66	1.13	0.91	0.95	0.99	1.01	1.15	1.05	1.29	1.55
ME-9	0.95	0.44	0.89	0.92	1.00	1.05	0.93	0.82	1.11	1.04	1.22
Large-ME	0.89	0.93	0.88	0.84	0.71	0.79	0.83	0.81	0.96	0.97	1.18

FF (1992): Evidence Based on Fama-MacBeth Regressions

	7/63–12/90 (330 Mos.)			7/63–12/76 (162 Mos.)			1/77–12/90 (168 Mos.)		
Variable	Mean	Std	t(Mn)	Mean	Std	t(Mn)	Mean	Std	t(Mn)
$R_{it} = a + b_{2,t}\ln(\mathsf{ME}_{it}) + b_{3,t}\ln(\mathsf{BE}/\mathsf{ME}_{it}) + e_{it}$									
a	1.77	8.51	3.77	1.86	10.10	2.33	1.69	6.67	3.27
b_2	-0.11	1.02	-1.99	-0.16	1.25	-1.62	-0.07	0.73	-1.16
b_3	0.35	1.45	4.43	0.36	1.53	2.96	0.35	1.37	3.30
$R_{it} = a + b_{1,t}\beta_{it} + b_{2,t}\ln(\mathsf{ME}_{it}) + b_{3,t}\ln(\mathsf{BE}/\mathsf{ME}_{it}) + e_{it}$									
a	2.07	5.75	6.55	1.73	6.22	3.54	2.40	5.25	5.92
b_1	-0.17	5.12	-0.62	0.10	5.33	0.25	-0.44	4.91	-1.17
b_2	-0.12	0.89	-2.52	-0.15	1.03	-1.91	-0.09	0.74	-1.64
b_3	0.33	1.24	4.80	0.34	1.36	3.17	0.31	1.10	3.67

FF (1992): Why "Death of CAPM"?

- CAPM is not rejected by one odd portfolio, but by broad return structure
- Beta does not span expected returns once size and value are present
- The average β_1 was in fact **negative** even though indistinguishable from zero
- The profession reads this as a model-level failure, not a small fix
- This was a *major* blow to the CAPM!

Questions?

Part II: From Characteristics to Factors

Motivation

- Fama and French (1992) showed that sorting stocks based on characteristics produced return patterns CAPM can't handle
- This kills CAPM... so what's next?

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- Fama and French (1992) showed that sorting stocks based on characteristics produced return patterns CAPM can't handle
- This kills CAPM… so what's next?
- Fama and French (1993) proposed a new factor model, meant to *substitute CAPM*
- A model based on 3 factors: the market, one size factor, and one book-to-market factor
- This is saying that your SDF moves as a function of the market + something else
- How to construct this "something else"? Mimicking portfolios!

The End Goal

Recall that a 3-factor model for (excess) returns implies:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1} F_{1,t} + \beta_{i,2} F_{2,t} + \beta_{i,3} F_{3,t} + \varepsilon_{i,t}$$

- Such model will price assets if $\alpha_i = 0$ for all $i \implies$ remember Ross (1976)!

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- Like the CAPM, this would imply that $R_{i,t} - R_{f,t}$ is only a function of factor exposures!
- A candidate for $F_{1,t}$ is the market factor, a proxy for "return on wealth"
- Fama and French (1993) constructed $F_{2,t}$ and $F_{3,t}$ to capture size and value patterns

SMB and HML Construction (Compact Version)

- They divide stocks, every June, into 6 groups along 2 dimensions:
  - Size: small (below median) and big (above median)
  - Book-to-market: low (bottom 30%), neutral (30-70%), and high (top 30%)
- Six final portfolios: SL_t, SN_t, SH_t, BL_t, BN_t, and BH_t

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Define the following:

$$\mathrm{SMB}_t = \frac{1}{3}(SL_t + SN_t + SH_t) - \frac{1}{3}(BL_t + BN_t + BH_t)$$

$$\mathrm{HML}_t = \frac{1}{2}(SH_t + BH_t) - \frac{1}{2}(SL_t + BL_t)$$

- SMB = small minus big returns, HML = high minus low book-to-market returns
- You can actually trade these portfolios

Three-Factor Regression and Alpha Restriction

They run the following time-series regression for each test asset i

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{iM}(R_{M,t} - R_{f,t}) + \beta_{iS}\,\mathrm{SMB}_t + \beta_{iH}\,\mathrm{HML}_t + \varepsilon_{i,t}$$

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What assets to test?

- They build 25 test portfolios
- These are 5x5 portfolios sorted on size and book-to-market
- Individual stocks used only to construct factors

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What assets to test?

- They build 25 test portfolios
- These are 5x5 portfolios sorted on size and book-to-market
- Individual stocks used only to construct factors
- What do you think is the argument against running this on individual stocks?

FF (1993): Restricted Model and Alpha Patterns

First, a restricted regression: $R_{i,t} - R_{f,t} = a_i + b_i[R_{M,t} - R_{f,t}] + e(t)$

	a					$t(a)$				
Size/ BE/ME	Low	2	3	4	High	Low	2	3	4	High
Small	-0.22	0.15	0.30	0.42	0.54	-0.90	0.73	1.54	2.19	2.53
2	-0.18	0.17	0.36	0.39	0.53	-1.00	1.05	2.35	2.79	3.01
3	-0.16	0.15	0.23	0.39	0.50	-1.12	1.25	1.82	3.20	3.19
4	-0.05	-0.14	0.12	0.35	0.57	-0.50	-1.50	1.20	2.91	3.71
Big	-0.04	-0.07	-0.07	0.20	0.21	-0.49	-0.95	-0.70	1.89	1.41

- Notice how $\hat{a}$ is monotonic!
- It increases with BE/ME and decreases with size

FF (1993): Key Figure Placeholder

The full model: $R_{i,t} - R_{f,t} = a_i + b_i[R_{M,t} - R_{f,t}] + s_i SMB_t + h_i HML_t + e_i(t)$

	a					$t(a)$				
Size/ BE/ME	Low	2	3	4	High	Low	2	3	4	High
Small	−0.34	−0.12	−0.05	0.01	0.00	−3.16	−1.47	−0.73	0.22	0.14
2	−0.11	−0.01	0.08	0.03	0.02	−1.24	−0.20	1.04	0.51	0.34
3	−0.11	0.04	−0.04	0.05	0.05	−1.42	0.47	−0.47	0.71	0.56
4	0.09	−0.22	−0.08	0.03	0.13	1.07	−2.65	−0.99	0.33	1.24
Big	0.21	−0.05	−0.13	−0.05	−0.16	3.27	−0.67	−1.46	−0.69	−1.41

- Estimated intercepts are much smaller and mostly insignificant
- In the paper: they also do the test by Gibbons, Ross, and Shanken (1989) The R^2 in the regressions is frequently above 90%

FF (1993): Some Comments

- They were able to construct portfolios that the CAPM couldn't price...
- But their new model could!
- Don't expect neat structural theories of why this worked

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- They were able to construct portfolios that the CAPM couldn't price...
- But their new model could!
- Don't expect neat structural theories of why this worked
- Taken at face value, there is no reason to use CAPM after this paper
- The 3-factor model should be used to estimate cost of capital, for example
- The alpha from the 3-factor model should be the metric for performance evaluation

Questions?

Part III: Momentum and the Interpretation Fight

Jegadeesh-Titman (1993): Momentum Design

- De Bondt and Thaler (1985): losers will be winners, and winners will be losers
- Horizon matters: past performance over 3-5 years, future performance over 3-5 years
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- Jegadeesh and Titman (1993): analyzed strategies based on shorter periods for returns: 3-12 months
- At time t: Evaluate post-formation returns over the following months
- Core result: return continuation is economically large

Jegadeesh-Titman (1993): Main Result

J	K =	3	6	9	12
3	Sell	0.0108	0.0091	0.0092	0.0087
		(2.16)	(1.87)	(1.92)	(1.87)
	Buy	0.0140	0.0149	0.0152	0.0156
		(3.57)	(3.78)	(3.83)	(3.89)
	Buy-sell	0.0032	0.0058	0.0061	0.0069
		(1.10)	(2.29)	(2.69)	(3.53)
6	Sell	0.0087	0.0079	0.0072	0.0080
		(1.67)	(1.56)	(1.48)	(1.66)
	Buy	0.0171	0.0174	0.0174	0.0166
		(4.28)	(4.33)	(4.31)	(4.13)
	Buy-sell	0.0084	0.0095	0.0102	0.0086
		(2.44)	(3.07)	(3.76)	(3.36)
9	Sell	0.0077	0.0065	0.0071	0.0082
		(1.47)	(1.29)	(1.43)	(1.66)
	Buy	0.0186	0.0186	0.0176	0.0164
		(4.56)	(4.53)	(4.30)	(4.03)
	Buy-sell	0.0109	0.0121	0.0105	0.0082
		(3.03)	(3.78)	(3.47)	(2.89)
12	Sell	0.0060	0.0065	0.0075	0.0087
		(1.17)	(1.29)	(1.48)	(1.74)
	Buy	0.0192	0.0179	0.0168	0.0155
		(4.63)	(4.36)	(4.10)	(3.81)
	Buy-sell	0.0131	0.0114	0.0093	0.0068
		(3.74)	(3.40)	(2.95)	(2.25)

- Winners minus losers is positive in almost all J-K combinations
- The effect is strongest for intermediate horizons (about 6-12 months)
- t-stats for buy-sell are often above 2, so not just noise
- This is hard to reconcile with a pure CAPM story

Isn't It Just Beta?

Portfolio (sorted on 6-month past returns)	Beta	Average Market Capitalization
P1	1.36	208.24
P2	1.19	480.07
P3	1.14	545.31
P4	1.11	618.85
P5	1.09	692.89
P6	1.08	702.51
P7	1.09	738.09
P8	1.12	758.87
P9	1.17	680.18
P10	1.28	495.13
P10–P1	−0.08	—

Average Returns By Past Performance Controlling For Beta and Size

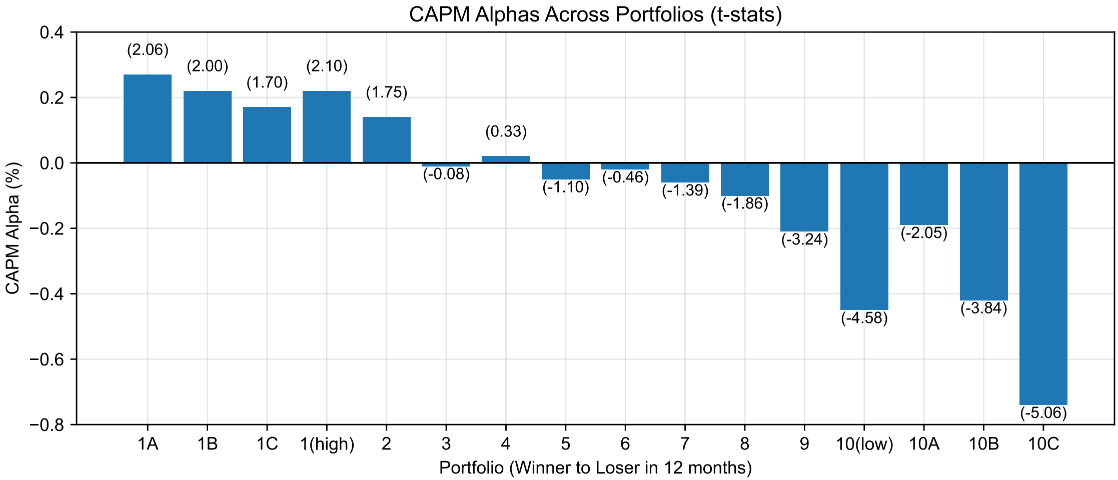
	All	S1	S2	S3	β_1	β_2	β_3
P1	0.0079	0.0083	0.0047	0.0082	0.0129	0.0097	0.0052
	(1.56)	(1.35)	(0.99)	(2.22)	(2.92)	(2.01)	(0.95)
P2	0.0112	0.0117	0.0102	0.0098	0.0140	0.0128	0.0086
	(2.78)	(2.29)	(2.54)	(3.08)	(4.38)	(3.37)	(1.83)
P3	0.0125	0.0152	0.0125	0.0105	0.0132	0.0133	0.0102
	(3.40)	(3.23)	(3.34)	(3.53)	(4.59)	(3.77)	(2.28)
P4	0.0124	0.0163	0.0130	0.0105	0.0134	0.0128	0.0110
	(3.59)	(3.59)	(3.58)	(3.66)	(5.02)	(3.82)	(2.50)
P5	0.0128	0.0164	0.0134	0.0109	0.0135	0.0135	0.0121
	(3.87)	(3.74)	(3.83)	(3.85)	(5.14)	(4.15)	(2.86)
P6	0.0134	0.0174	0.0146	0.0102	0.0135	0.0142	0.0122
	(4.14)	(4.08)	(4.22)	(3.66)	(5.23)	(4.38)	(2.92)
P7	0.0136	0.0175	0.0143	0.0109	0.0136	0.0142	0.0126
	(4.19)	(4.13)	(4.12)	(3.90)	(5.09)	(4.43)	(3.01)
P8	0.0143	0.0174	0.0148	0.0111	0.0143	0.0146	0.0132
	(4.30)	(4.11)	(4.16)	(3.86)	(5.12)	(4.44)	(3.15)
P9	0.0153	0.0183	0.0154	0.0126	0.0165	0.0156	0.0141
	(4.36)	(4.28)	(4.11)	(4.17)	(5.34)	(4.56)	(3.28)
P10	0.0174	0.0182	0.0173	0.0157	0.0191	0.0176	0.0160
	(4.33)	(3.99)	(4.11)	(4.41)	(5.17)	(4.53)	(3.50)
P10–P1	0.0095	0.0099	0.0126	0.0075	0.0062	0.0079	0.0108
	(3.07)	(2.77)	(4.57)	(3.03)	(2.05)	(2.64)	(3.35)
F-Statistics[a]	2.83	2.65	4.51	4.38	2.51	1.99	1.69
p-Value	(0.00)	(0.00)	(0.00)	(0.00)	(0.01)	(0.04)	(0.09)

Carhart (1997): Four-Factor Extension

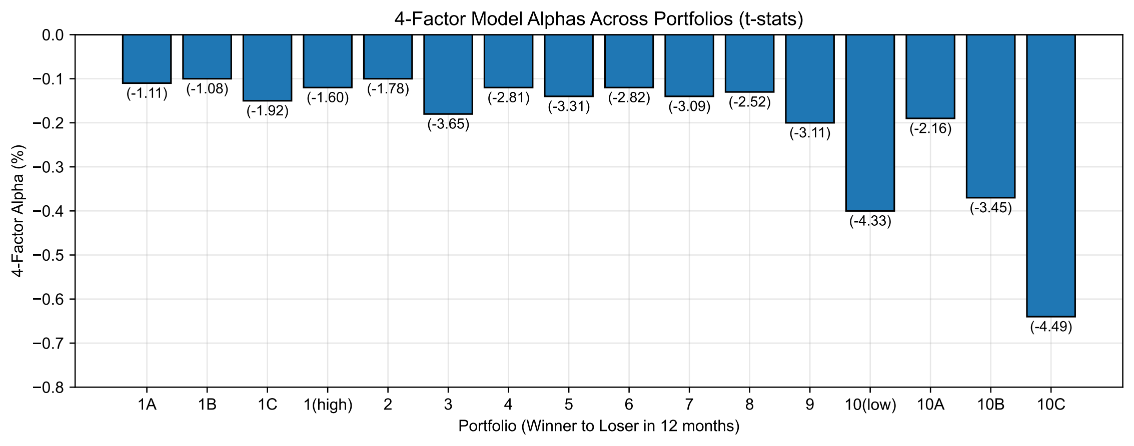
$$R_{i,t} - R_{f,t} = \alpha_i + \beta_M MKT_t + \beta_S SMB_t + \beta_H HML_t + \beta_U UMD_t + \varepsilon_{i,t}$$

- Add a momentum factor (UMD) to FF3
- UMD stands for "up minus down"
- Constructed as a long-short portfolio based on past 12-month returns
- Up = top 30% winners, down = bottom 30% losers
- This four-factor model became the standard mutual-fund benchmark
- It is mainly an empirical benchmark, not a full structural theory

CAPM Alphas: 1A = highest past-year returns



Four-Factor Model Alphas: 1A = highest past-year returns



Carhart (1997): Why Are Funds So Bad?

For each individual fund, compute:

$$\alpha_{i,t} = R_{i,t} - R_{f,t} - \hat{\beta}_{M,i}(R_{M,t} - R_{f,t}) - \hat{\beta}_{S,i}SMB_t - \hat{\beta}_{H,i}HML_t - \hat{\beta}_{U,i}UMD_t$$

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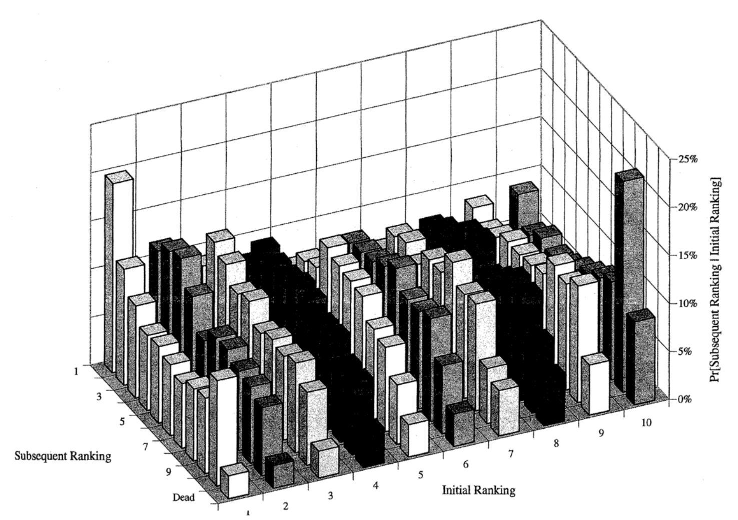
For any characteristic $x_{i,t}$, regress $\alpha_{i,t} = a + bx_{i,t} + \varepsilon_{i,t}$

Independent Variables	Estimate	t-statistic
Expense ratio (t)	-1.54	(-5.99)
Turnover (t) (Mturn)	-0.95	(-2.36)
$\ln$ TNA (t-1)	-0.05	(-0.66)
Maximum Load (t-1)	-0.11	(-3.55)
Buy turnover (t)	-0.43	(-1.16)
Sell turnover (t)	-1.26	(-3.00)

Carhart Became the Benchmark

- It captures momentum that FF3 leaves behind
- It is simple, tradable, and easy to implement in regressions
- It improved performance evaluation for funds and strategies
- Terrible funds stay terrible, great funds stay great, but everyone else is kind of random
- It still leaves interpretation open: risk compensation or mispricing?

Contingency Matrix



Key Takeaways from Carhart (1997)

- Most mutual funds destroy value, especially the very expensive ones
- Avoid funds with poor track record
- Measure performance against the 4-factor model, not CAPM
- Fees and transaction costs matter a lot
- Cynical take: work for a mutual fund, but save through an ETF

Questions?

Part IV: Is CAPM Really Dead? Or Just Sleeping?

Day vs Night Returns

- CAPM tests and everything we did typically use monthly returns
- You can do everything at daily frequency too...
- And even intraday! Hendershott, Livdan, and Rösch (2020) is a fun paper to read!

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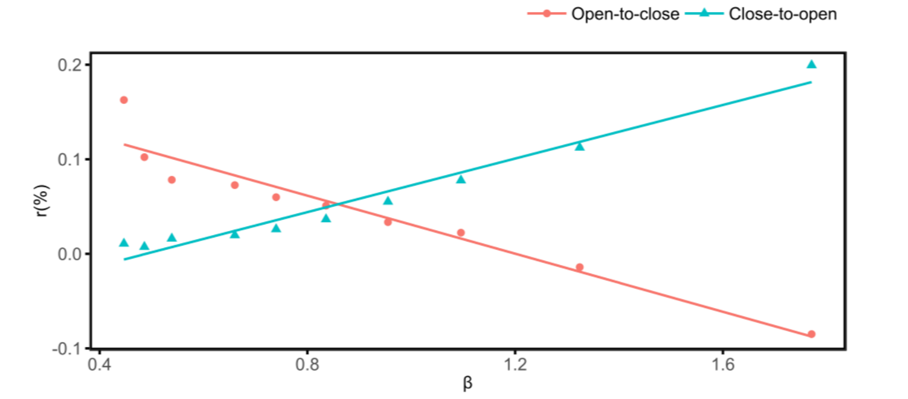
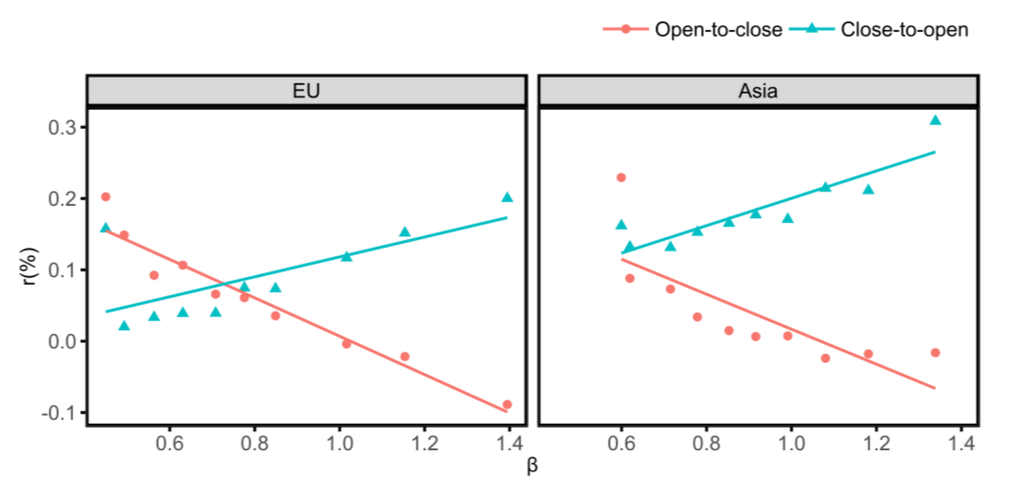
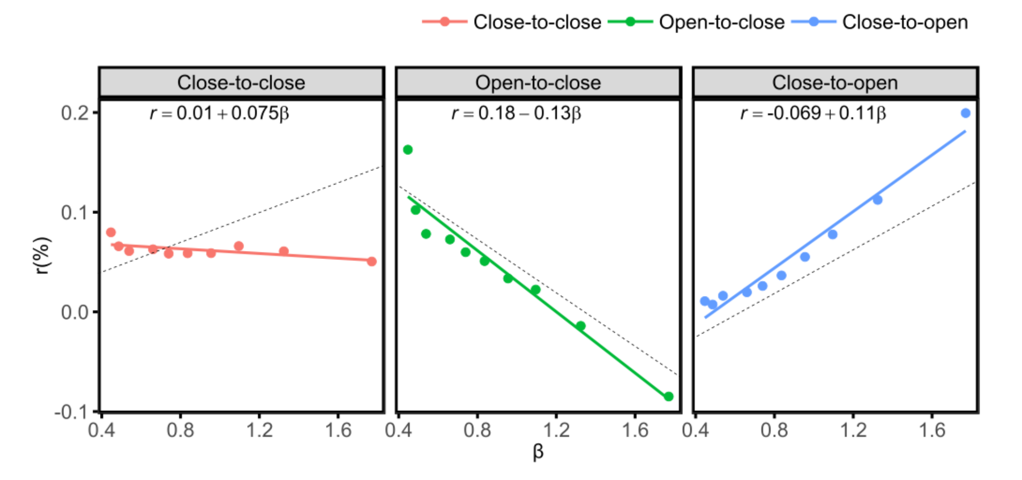


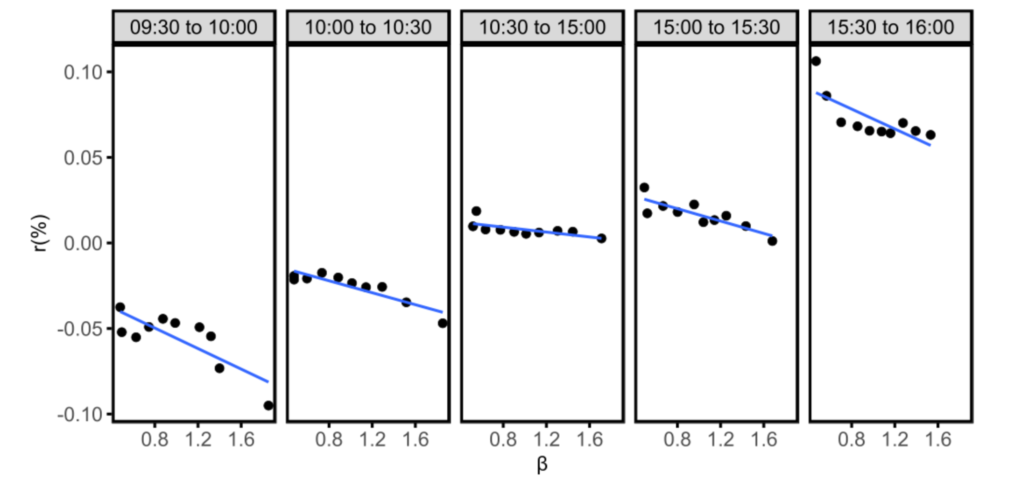
This is not only in the US



Different Timings



Intraday



09:30 to 10:00
10:00 to 10:30
10:30 to 15:00
15:00 to 15:30
15:30 to 16:00
0.10
0.05
0.00
-0.05
-0.10
r(%)
0.8
1.2
1.6
β

Wrap-Up

- Fama and French (1992) showed that CAPM fails to price returns sorted on size and book-to-market
- Fama and French (1993) proposed a 3-factor model that can price those returns
- Jegadeesh and Titman (1993) documented momentum in returns, which is not captured by CAPM or FF3
- Carhart (1997) added a momentum factor to FF3, creating the 4-factor model
- Hendershott, Livdan, and Rösch (2020): it's only really sleepy!

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Readings and References ii

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